

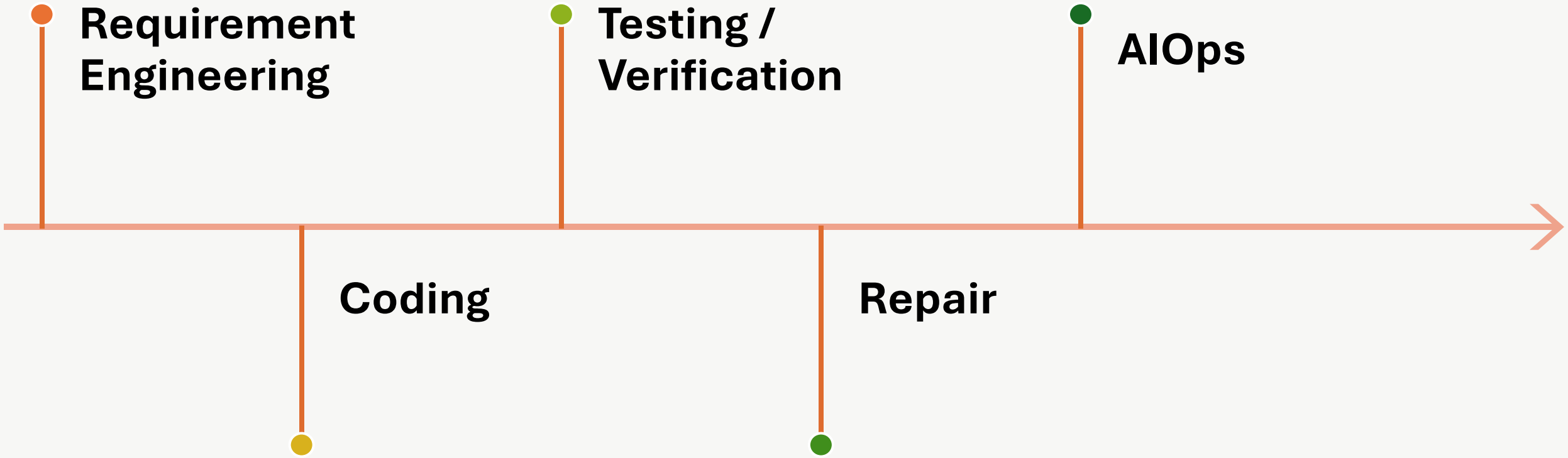
香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

CSC4801
AI-assisted Software Engineering

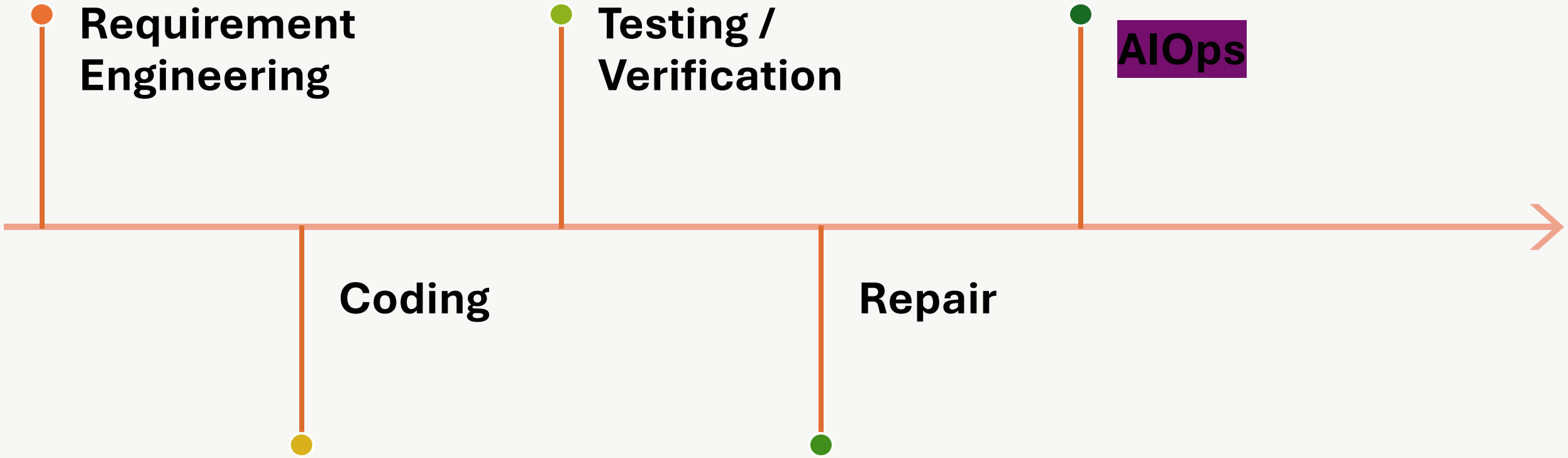
Lecture 17: AIOps

Instructor: Jinsheng Ba | 2026 Fall

AI-Assisted Software Engineering

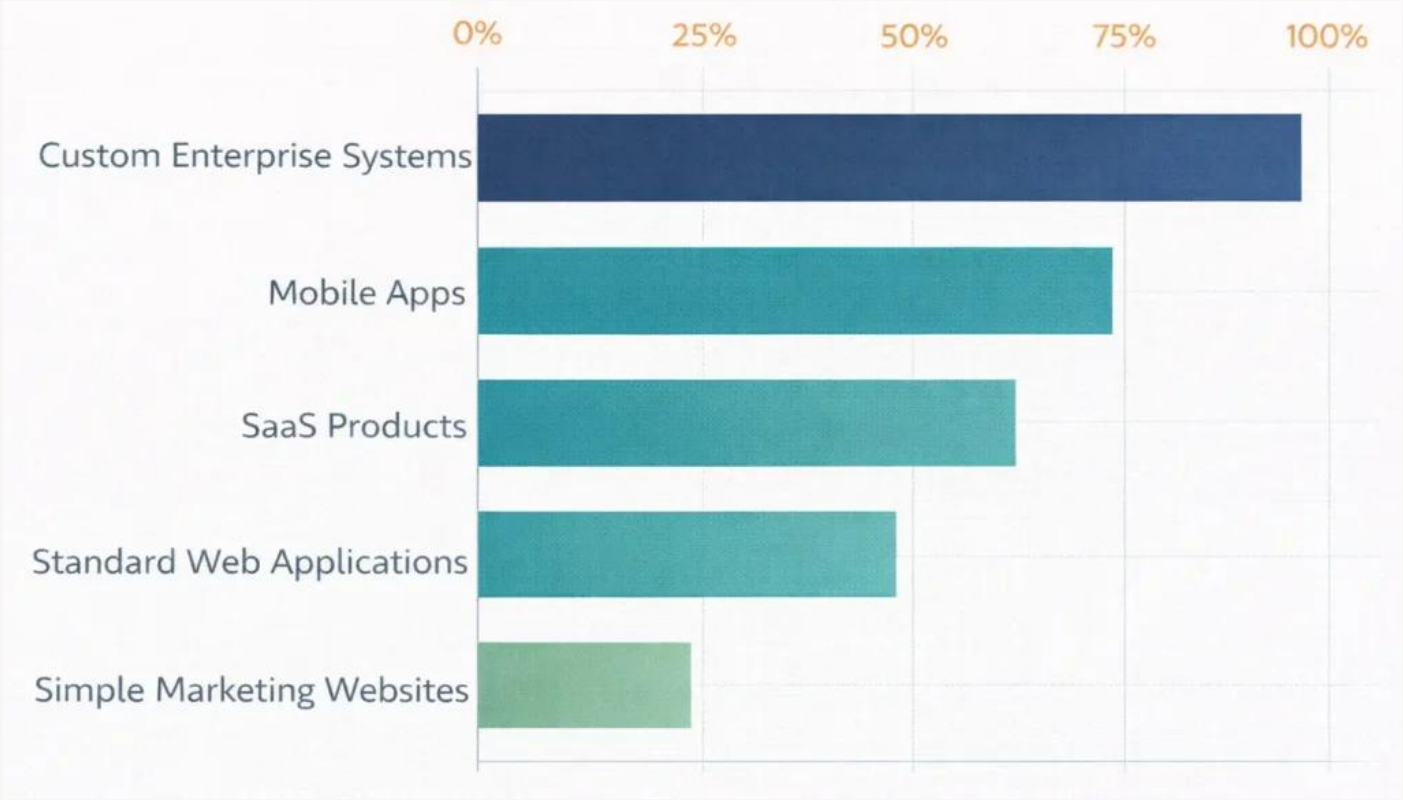


AI-Assisted Software Engineering

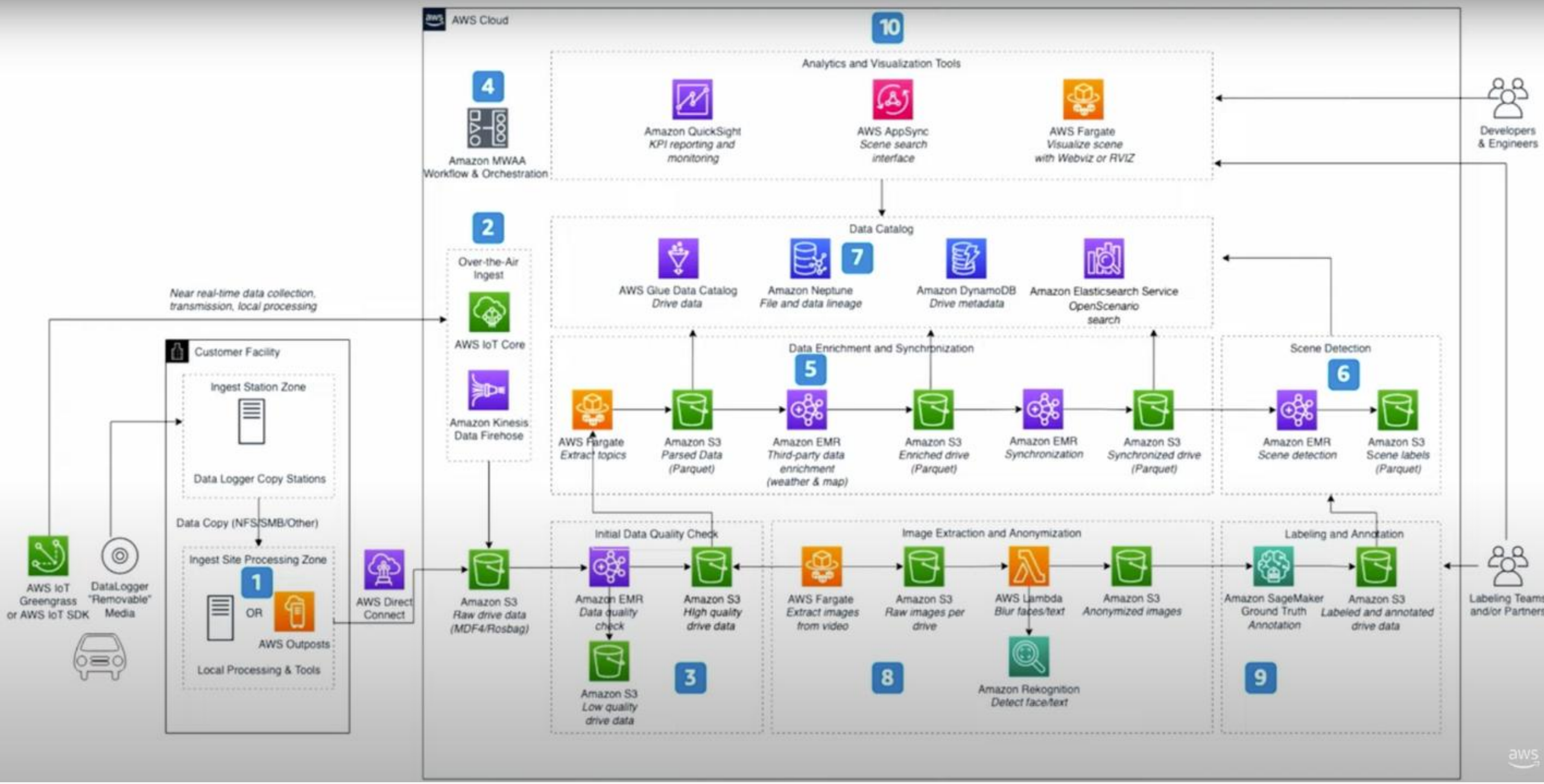


Software Maintenance

Gartner reports that organizations spend 55 to 80 percent of their IT budgets on maintaining existing systems.



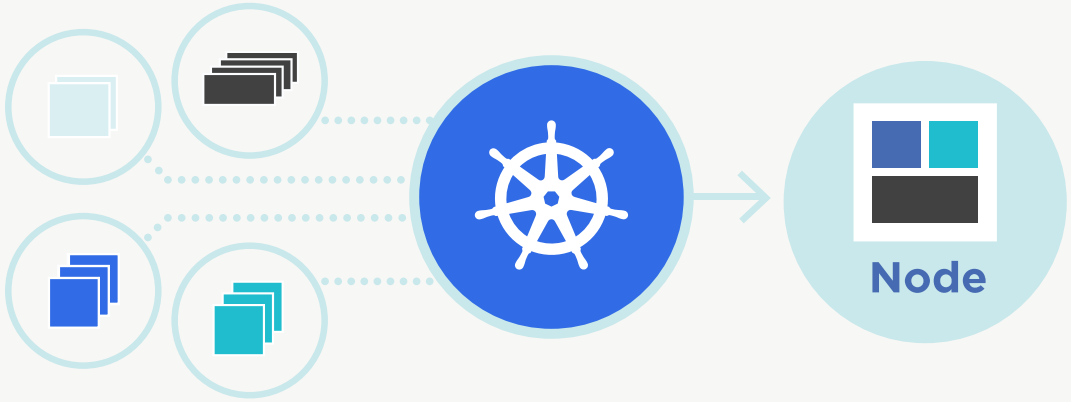
https://pegotec.net/software-maintenance-cost-percentage-2026-industry-benchmarks/



https://aws.amazon.com/blogs/architecture/lets-architect-architecture-tools/

Before LLMs

- Site Reliability Engineering (SRE)
  - On-call for troubleshooting
- Cloud-native architectures (Kubernetes)
  - Partial automatic



AI DevOps

Principles of Infrastructure and DevOps

- Four golden signals of monitoring
  - Latency
  - Errors
  - Traffic
    - Requests/sec
  - Saturation
- Monitor production traces

At 3 am, you get a ping that you're seeing a spike in 500s on your database queries. What do we do?

A Potential Playbook

- Acknowledge and assess
- Check DB + app
- Identify recent changes
- Localize blast radius
- Execute mitigations
- Stabilize + monitor
- Communicate
- Document



https://www.linkedin.com/posts/sylvainkalache_there-are-only-2-engineers-who-never-get-activity-7447598621438500864-kVKQ/

Metrics Tracked

- Mean-time-to-repair (MTTR)
- How many engineers are pulled into incident

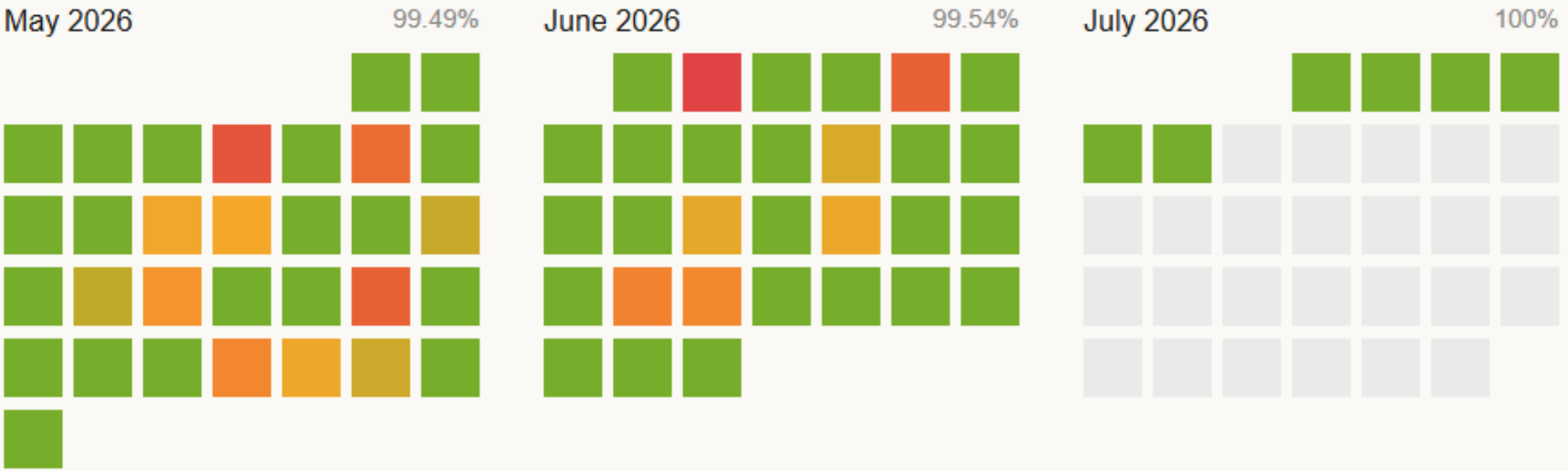
Claude Status

SUBSCRIBE TO UPDATES

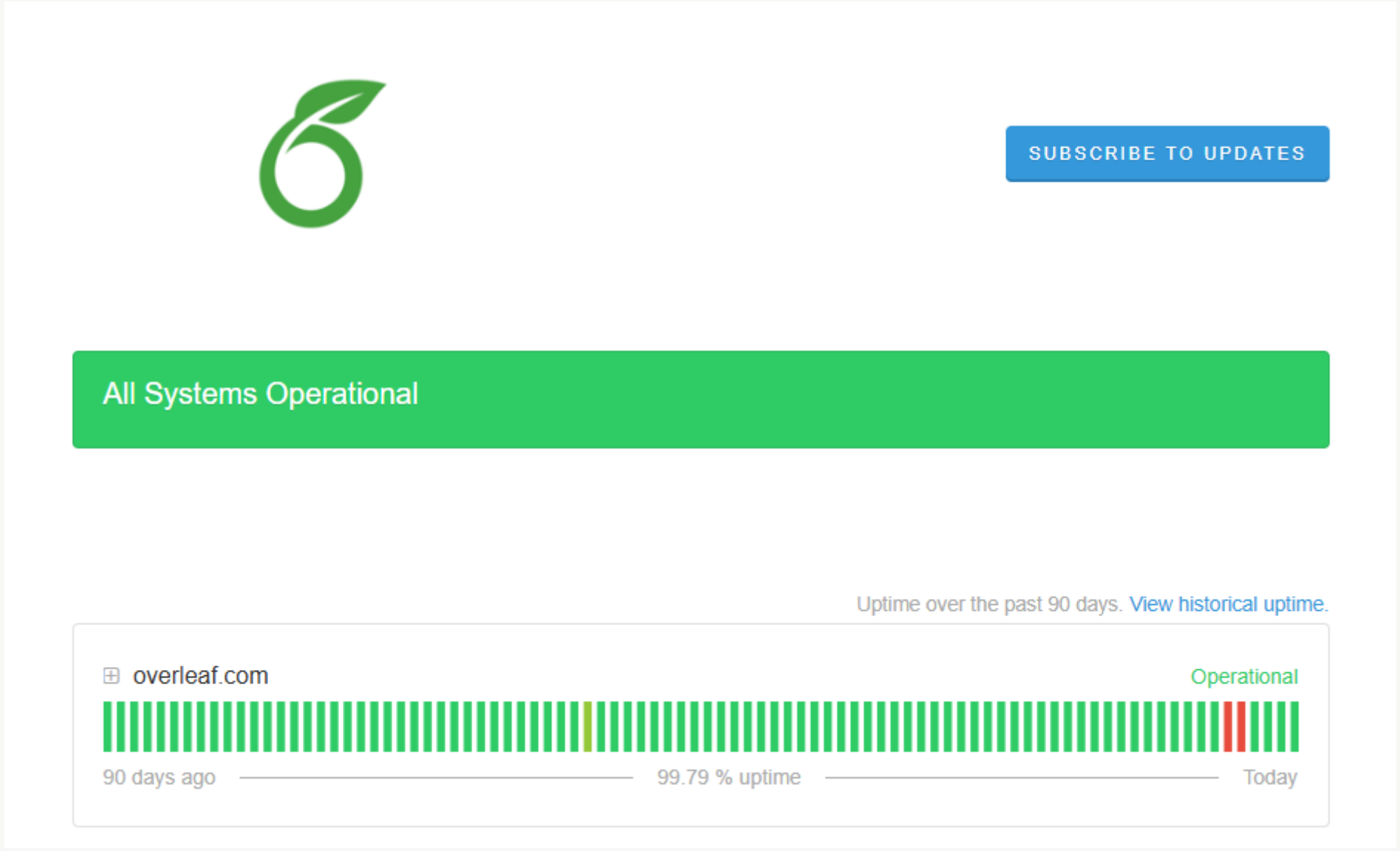
Incidents | Uptime

claude.ai

May 2026 to July 2026



https://status.claude.com/uptime



https://status.overleaf.com/

Three Tasks in AIOps



Failure Perception

Detecting whether anomalies have occurred in the system.

Data Source

- Metric
  - CPU usage, memory utilization, disk I/O.
- Trace
  - The sequence of operations or transactions that a request undergoes.
- Logs
  - Error messages, transaction records, user activities.

Logs

Logging Statements

```
# Logging statements from Spark (spark/storage/BlockManager.scala)
private def reportAllBlocks(): Unit = {
    logInfo(s"Reporting ${blockInfoManager.size} blocks to the master.")
    for ((blockId, info) <- blockInfoManager.entries) {
        // ...
        if (info.tellMaster && !tryToReportBlockStatus(blockId, status))
            logError(s"Failed to report $blockId to master; giving up.")
            // ...
        }
    }
}
```



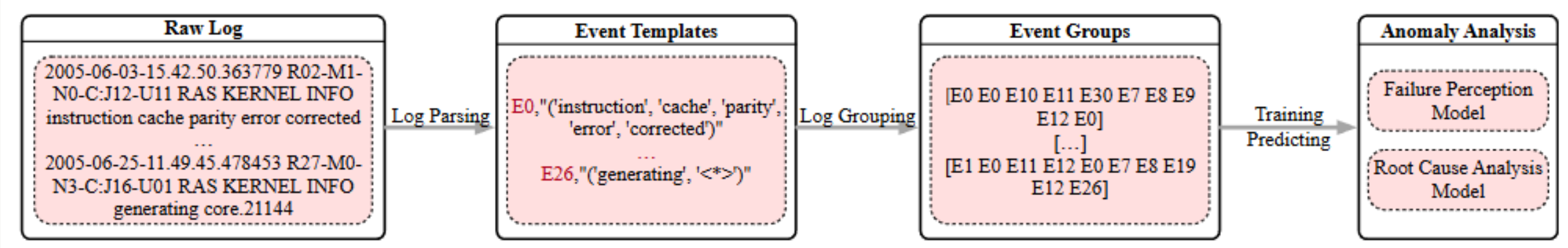
Log Message

```
17/08/22 15:50:02 INFO BlockManager Reporting 12 blocks to the master.
17/08/22 15:50:46 ERROR BlockManager Failed to report rdd_5_1 to master; giving up.
17/08/22 15:52:36 ERROR BlockManager Failed to report rdd_3_3 to master; giving up.
...
```

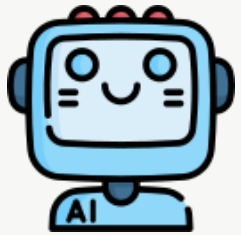
https://dl.acm.org/doi/pdf/10.1145/3654966

Logs

A common workflow



https://arxiv.org/pdf/2507.12472



AI infers log templates for parsing.

Prompt

```
Extract one log template, \
substitute variable tokens in the log as <*> between <START> and <END> tags.


Message: Recalculating schedule, headroom=<memory:0, vCores:-27>
Template: <START> Recalculating schedule, headroom=<memory:<*>, vCores:<*>> <END>

Message: Progress of TaskAttempt attempt_000001_0 is : 0.052178
Template: <START> Progress of TaskAttempt attempt_<*> is : <*> <END>

Message: task_000002 Task Transitioned from SCHEDULED to RUNNING
Template: <START> task_<*> Task Transitioned from SCHEDULED to RUNNING <END>

Message: attempt_000001 TaskAttempt Transitioned from ASSIGNED to RUNNING
Template: <START> attempt_<*> TaskAttempt Transitioned from ASSIGNED to RUNNING <END>

Message: JVM with ID : jvm_000003 asked for a task
Template: <START> JVM with ID : jvm_<*> asked for a task <END>


Message: JVM with ID: jvm_000005 given task: attempt_000003_0
```

Raw Output

```
<START> JVM with ID: jvm_<*> given task: attempt_<*> <END>
          ...
          ...
(Other Redundant Text Contents)
```

Filtered Output

```
JVM with ID: jvm_<*> given task: attempt_<*>
```

https://arxiv.org/pdf/2307.09950v3

More Data Source

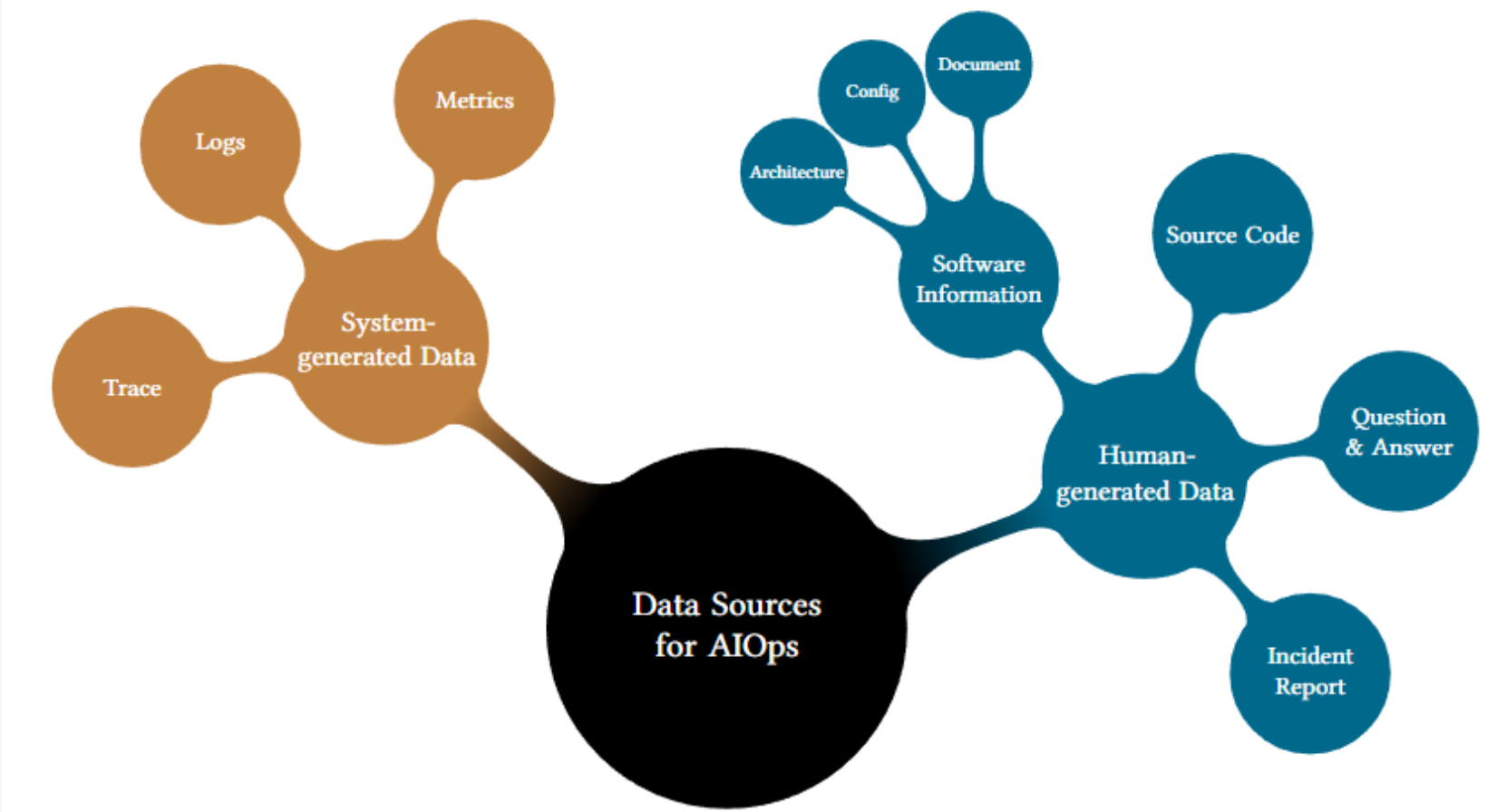
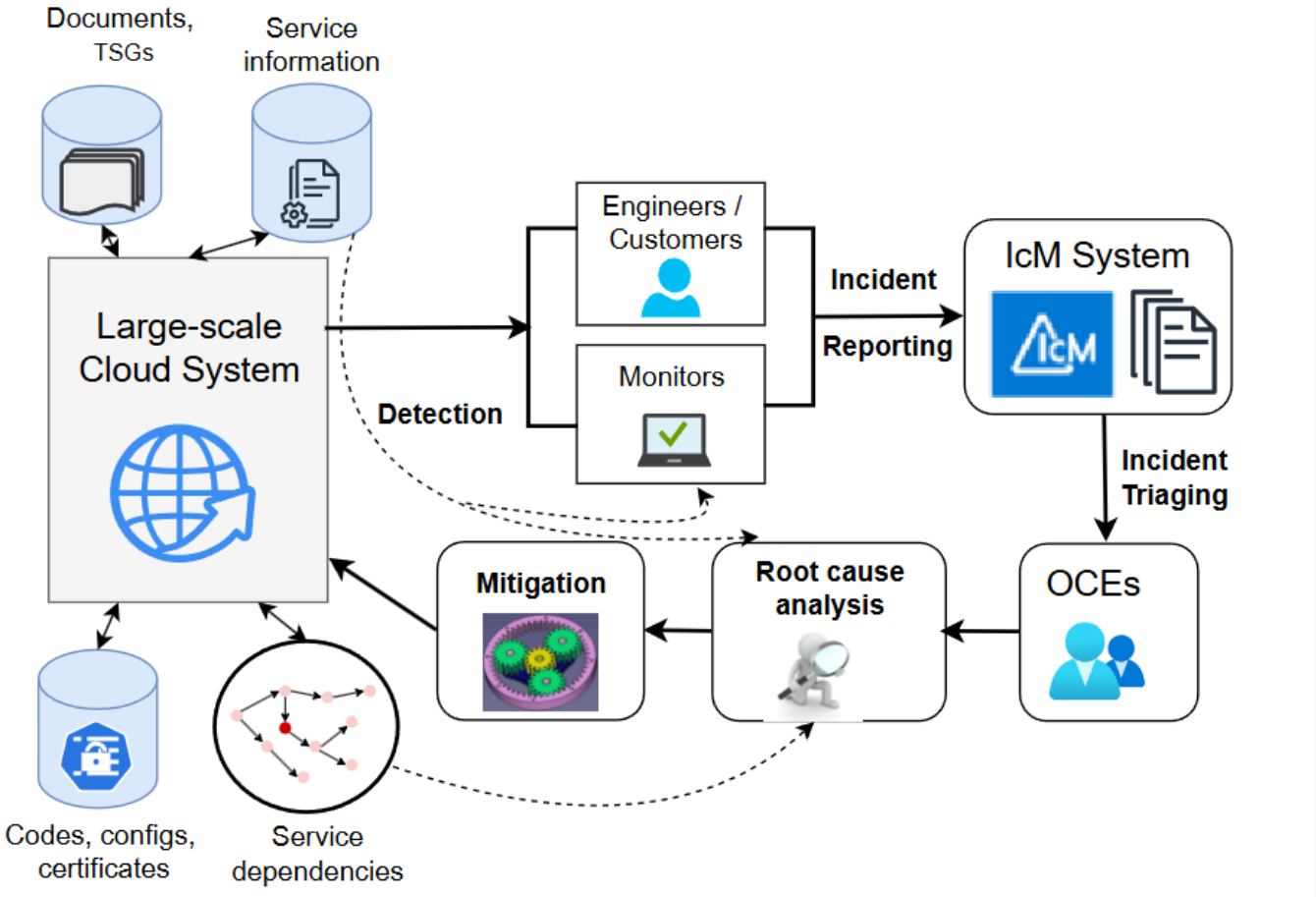


Fig. 6. Data Source for AIOps in the Era of LLM

https://arxiv.org/pdf/2507.12472

Dependent-Service Descriptions



https://arxiv.org/pdf/2507.12472

Dependent-Service Descriptions

I want you to act as an expert software engineer. Consider the service descriptions.

Upstream service 1 - Description

. . .

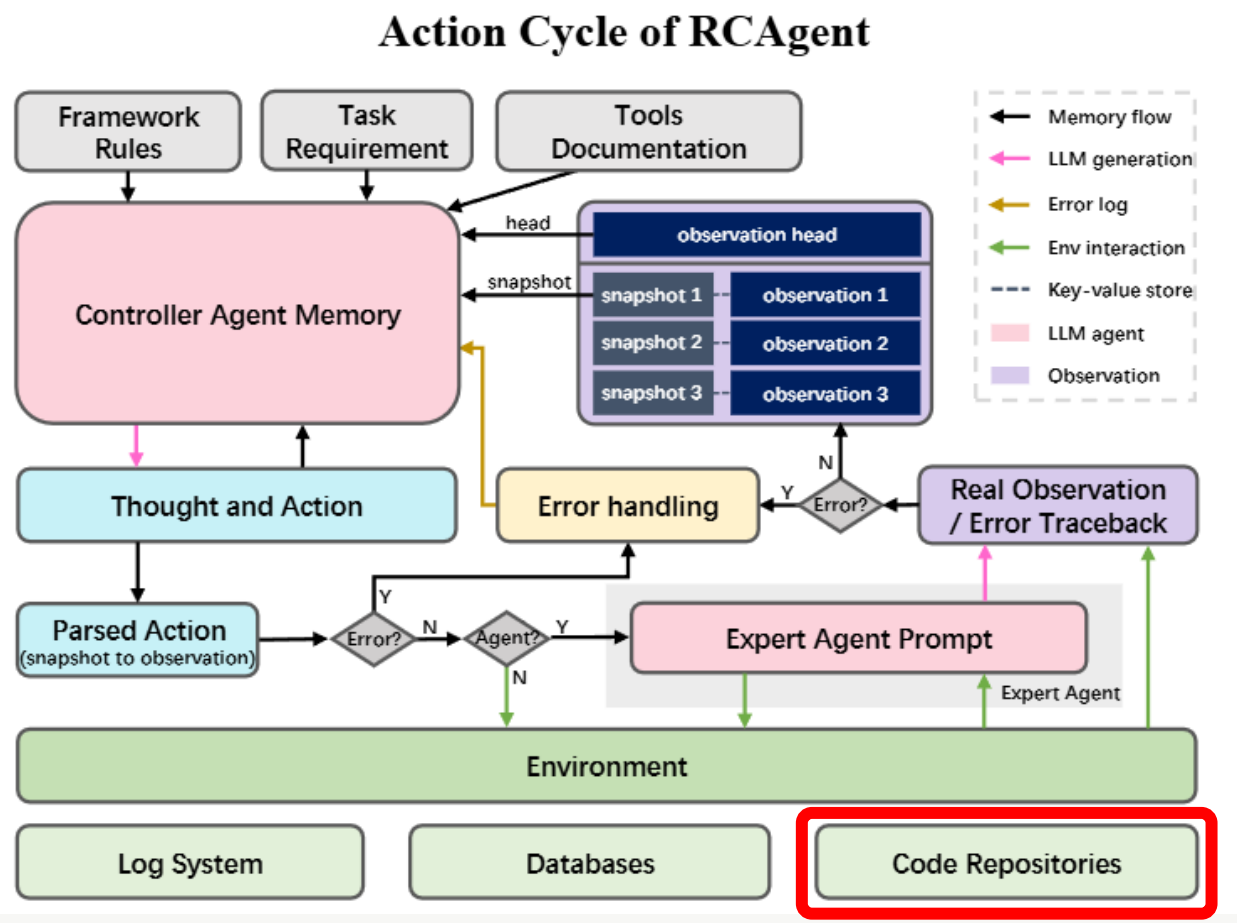
Upstream service n - Description

Your task is to summarize these service descriptions.
Focus on the following aspects of the service functionality:
* The functionality provided by the service
* References to resources it operates
* Distinguishing features of the service that clearly explains its operation
Your summary should be at most 2-3 sentences and should be in third person.

Figure 4: Prompt to summarize the Service Descriptions

https://arxiv.org/pdf/2507.12472

Source Code



https://arxiv.org/pdf/2310.16340

Failure Perception

Use historical logs to fine-tune LLMs for next event prediction.



Fig. 1: Module 1: Log message preprocessing

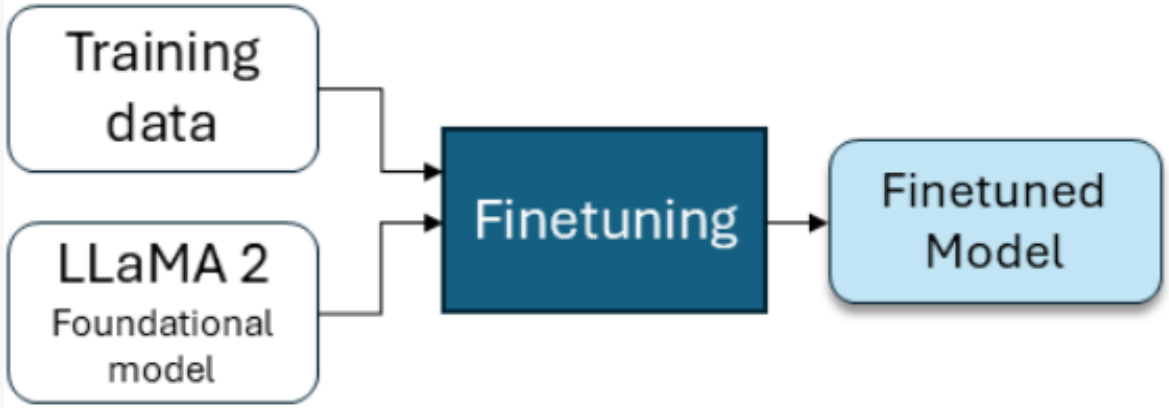


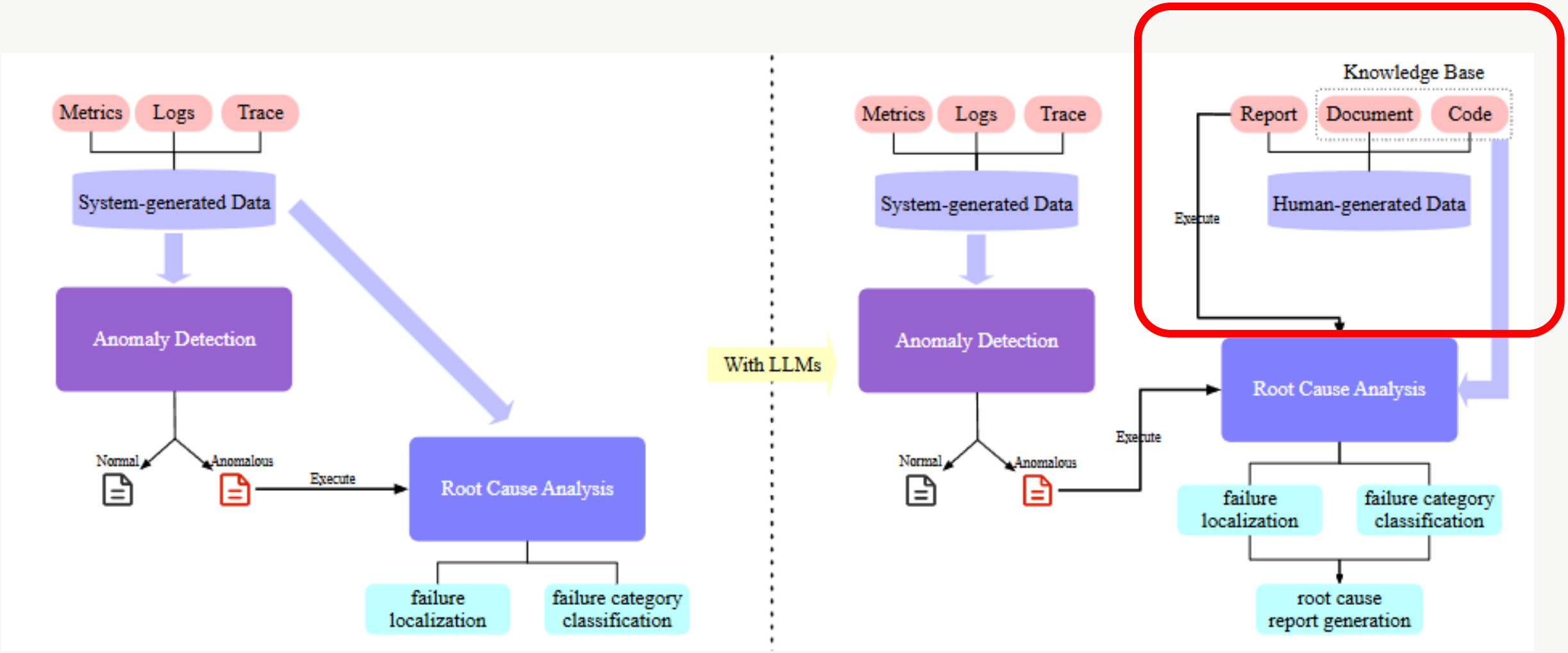
Fig. 2: Module 2: Model finetuning

https://arxiv.org/pdf/2503.14849v1

Three Tasks in AIOps



Root-Cause Analysis w/o LLMs



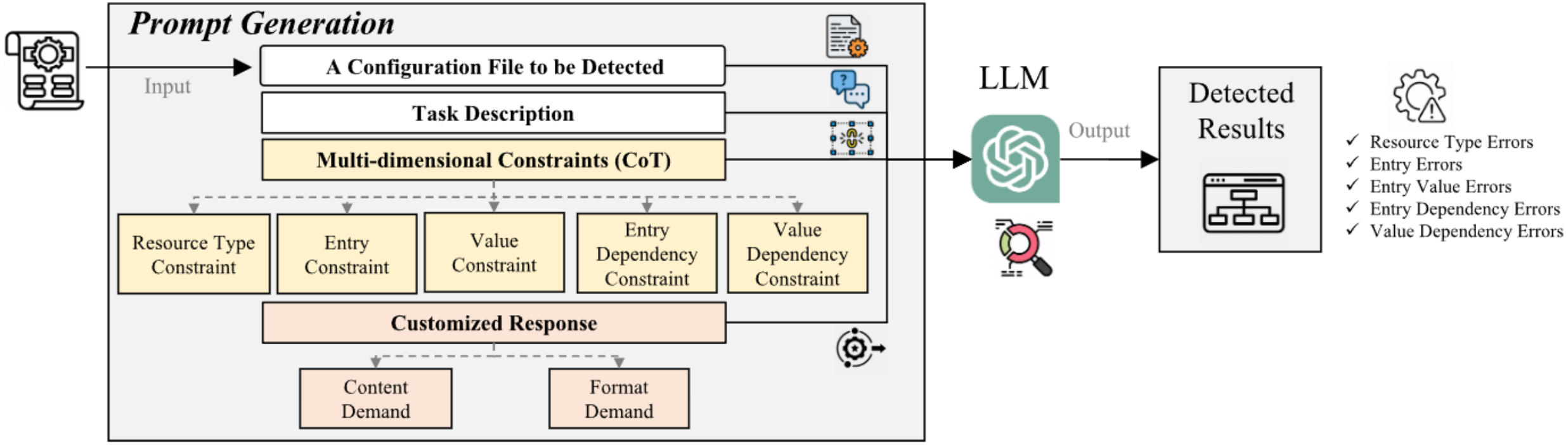
https://arxiv.org/pdf/2507.12472

Failure Localization

Identify the specific component or machine where an anomaly occurred

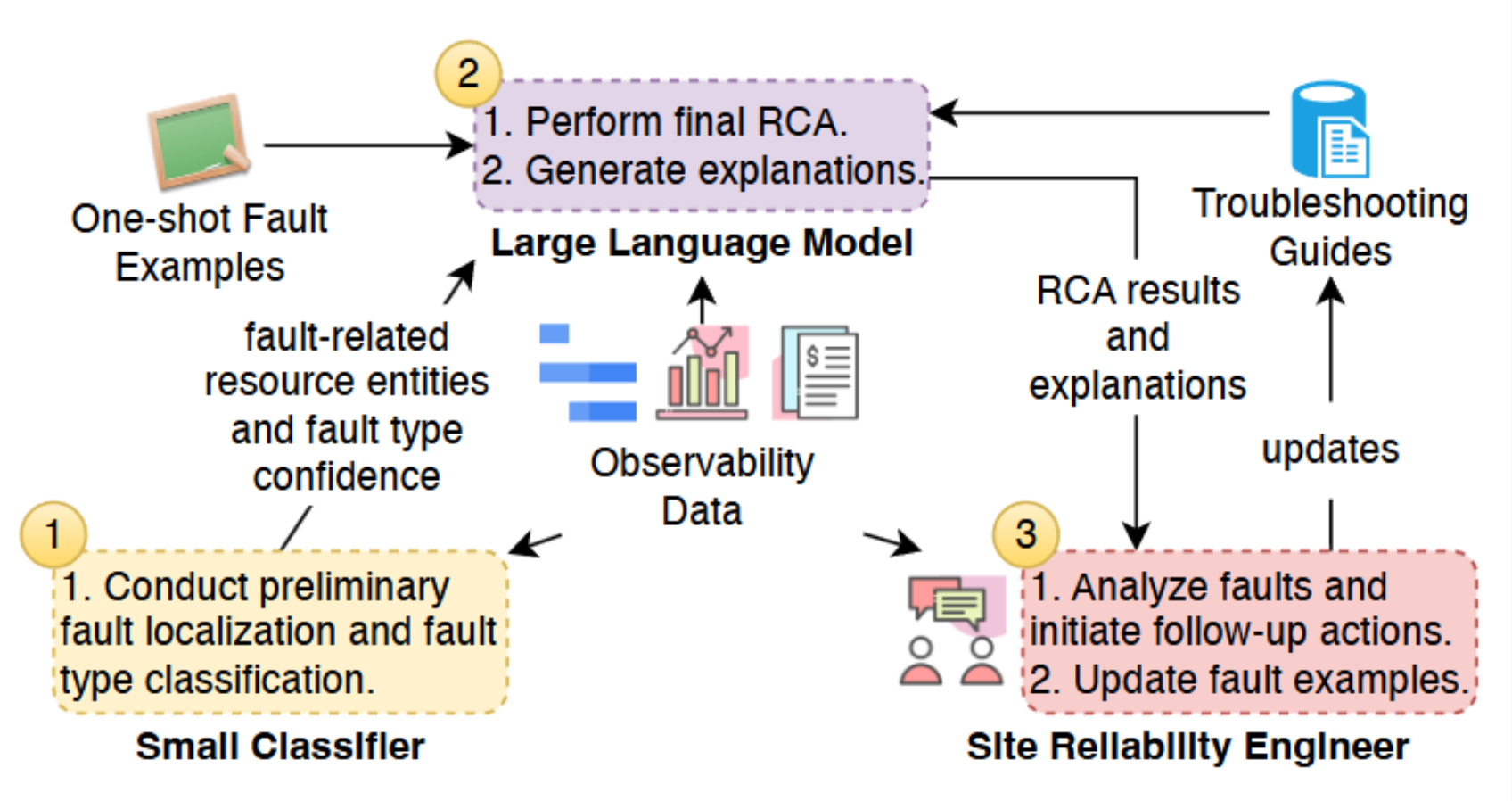
Prompting Engineering: CoT

Static scanning for misconfigurations



https://arxiv.org/pdf/2411.00642

Prompting Engineering: One-Shot

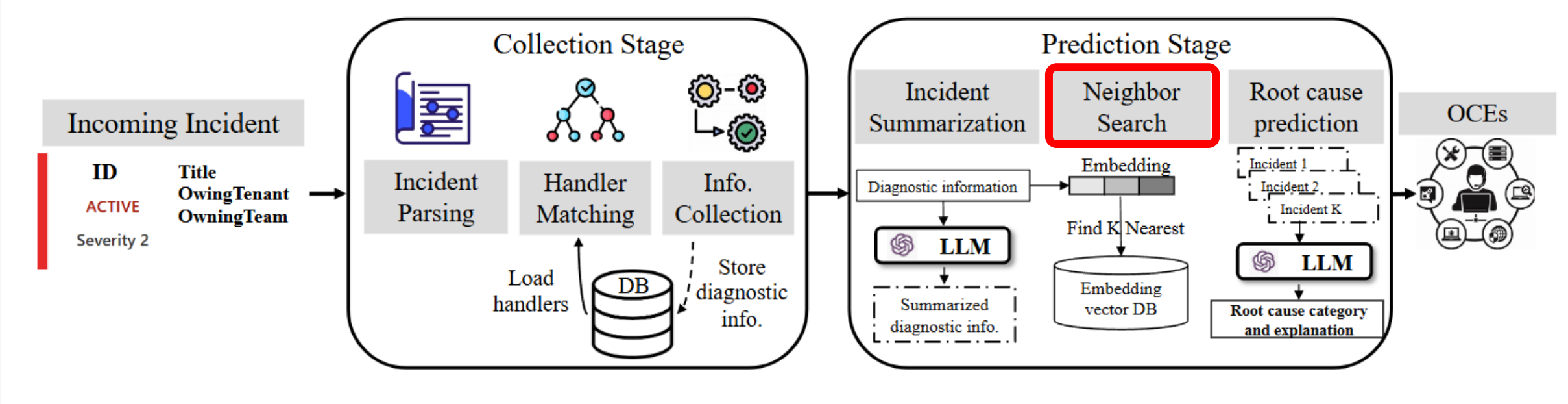


https://dl.acm.org/doi/pdf/10.1145/3691620.3695475

Failure Category Classification

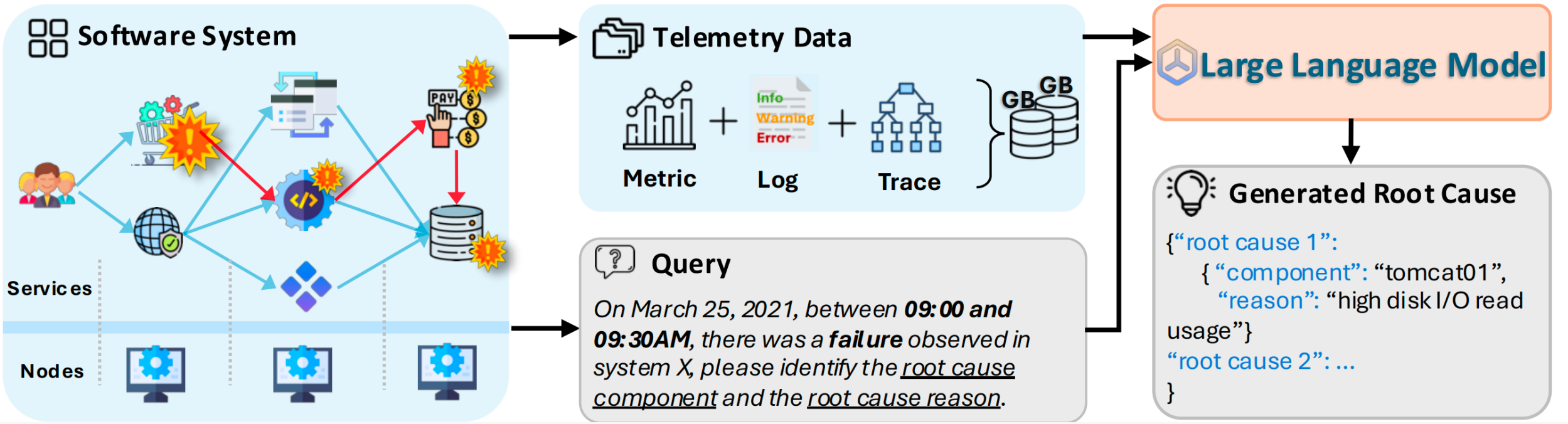
Determine the type of anomaly the system is experiencing.

Similarity Search



https://arxiv.org/pdf/2305.15778

Can LLMs do Root Cause Analysis?



https://microsoft.github.io/OpenRCA/

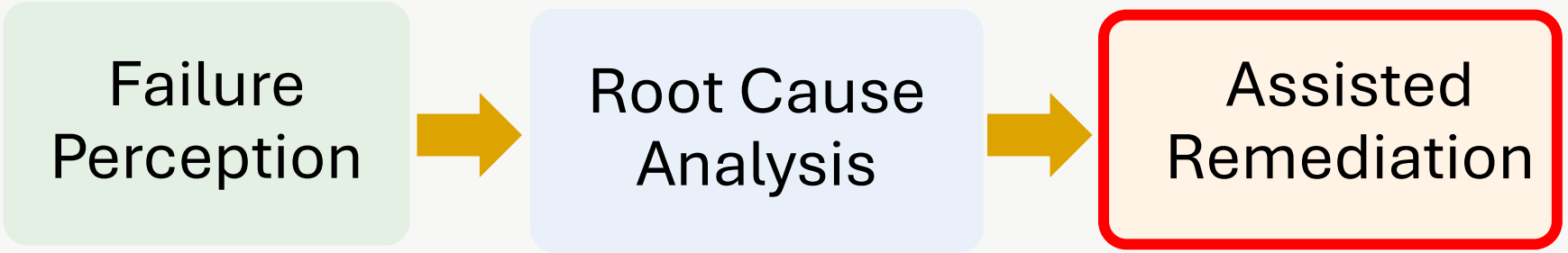
Can LLMs do Root Cause Analysis?

Method Name	Model	f1	acc ↓	node-F1	edge-F1	any-hit	all-hit	path-acc	type-acc
DeepResearch	Claude Fable 5	53.36%	35.20%	78.80%	66.07%	85.20%	51.20%	62.00%	73.59%
DeepResearch	GPT-5.5	41.57%	26.80%	77.26%	62.77%	84.80%	51.40%	47.80%	57.55%
DeepResearch	GPT-5.4	40.83%	25.80%	74.48%	59.42%	84.20%	50.40%	46.60%	57.48%
DeepResearch	Claude Opus 4.6	40.40%	25.20%	77.34%	62.29%	82.00%	48.80%	47.40%	58.41%
DeepResearch	Kimi K2.6	39.60%	24.80%	74.11%	58.18%	80.40%	48.00%	45.89%	58.46%
DeepResearch	Claude Opus 4.8	39.23%	24.00%	74.55%	56.44%	80.60%	46.40%	46.40%	58.19%
DeepResearch	Gemini 3.5 Flash	38.63%	23.60%	70.05%	54.29%	77.60%	45.20%	42.80%	59.41%
DeepResearch	DeepSeek V4 Pro	36.83%	23.60%	71.37%	54.90%	78.60%	47.60%	42.00%	55.47%
DeepResearch	Gemini 3.1 Pro	37.47%	23.40%	73.55%	57.43%	80.40%	47.60%	43.00%	55.22%

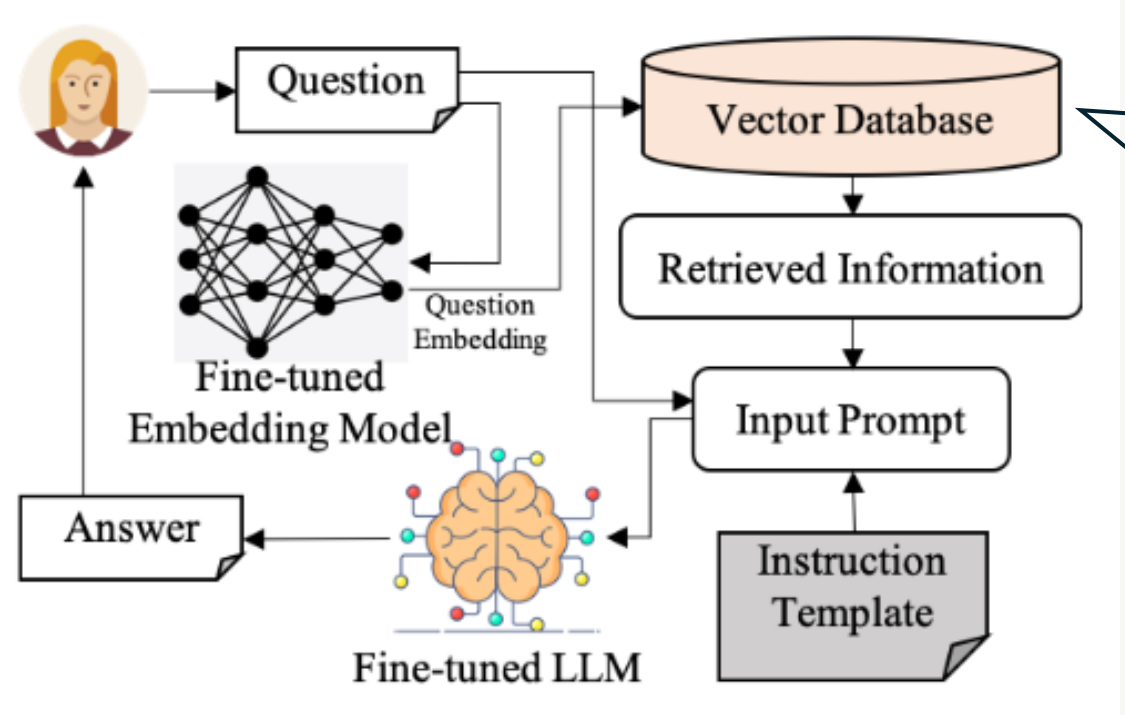
Note: All data is collected in China. Please use **UTC+8** timezone when converting timestamps to datetime.

https://microsoft.github.io/OpenRCA/

Three Tasks in AIOps



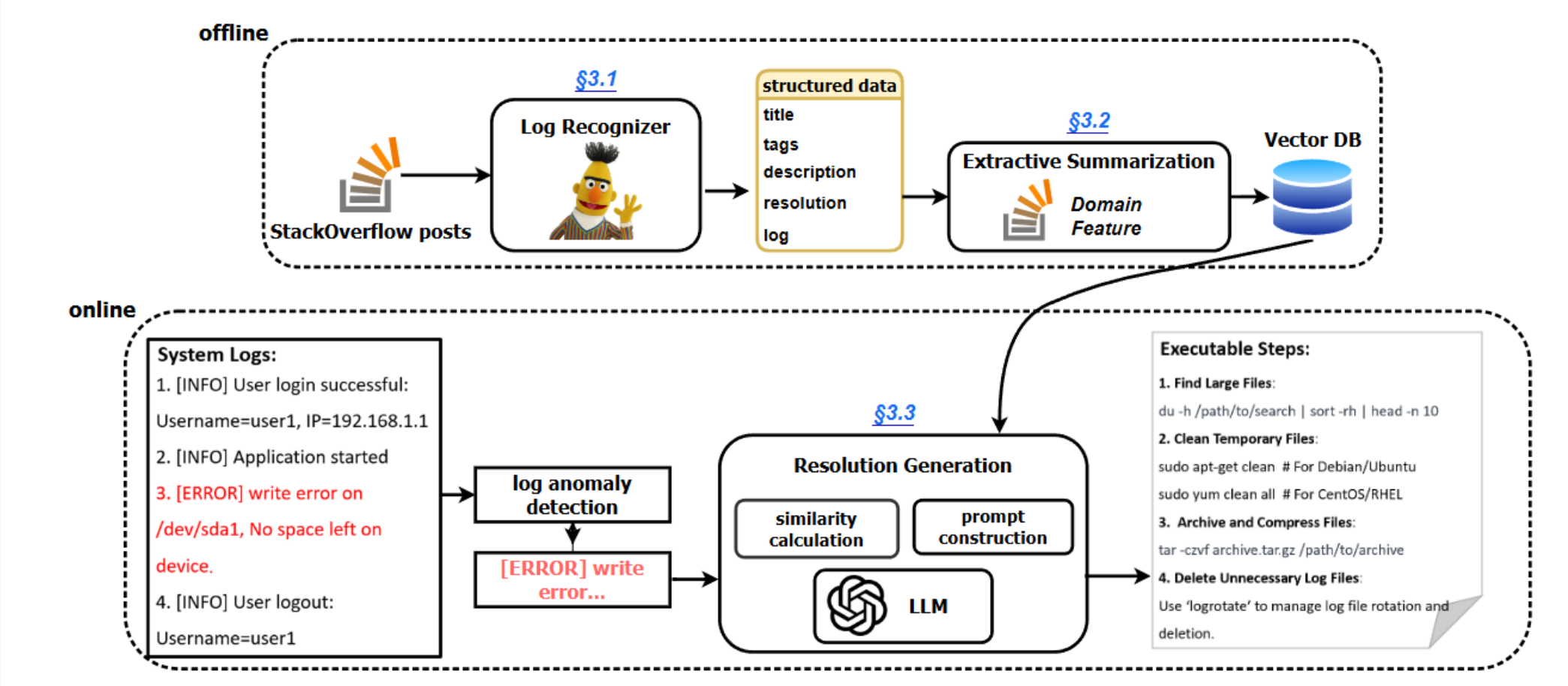
Incidents to Q/A



Summarizes incidents into databases.

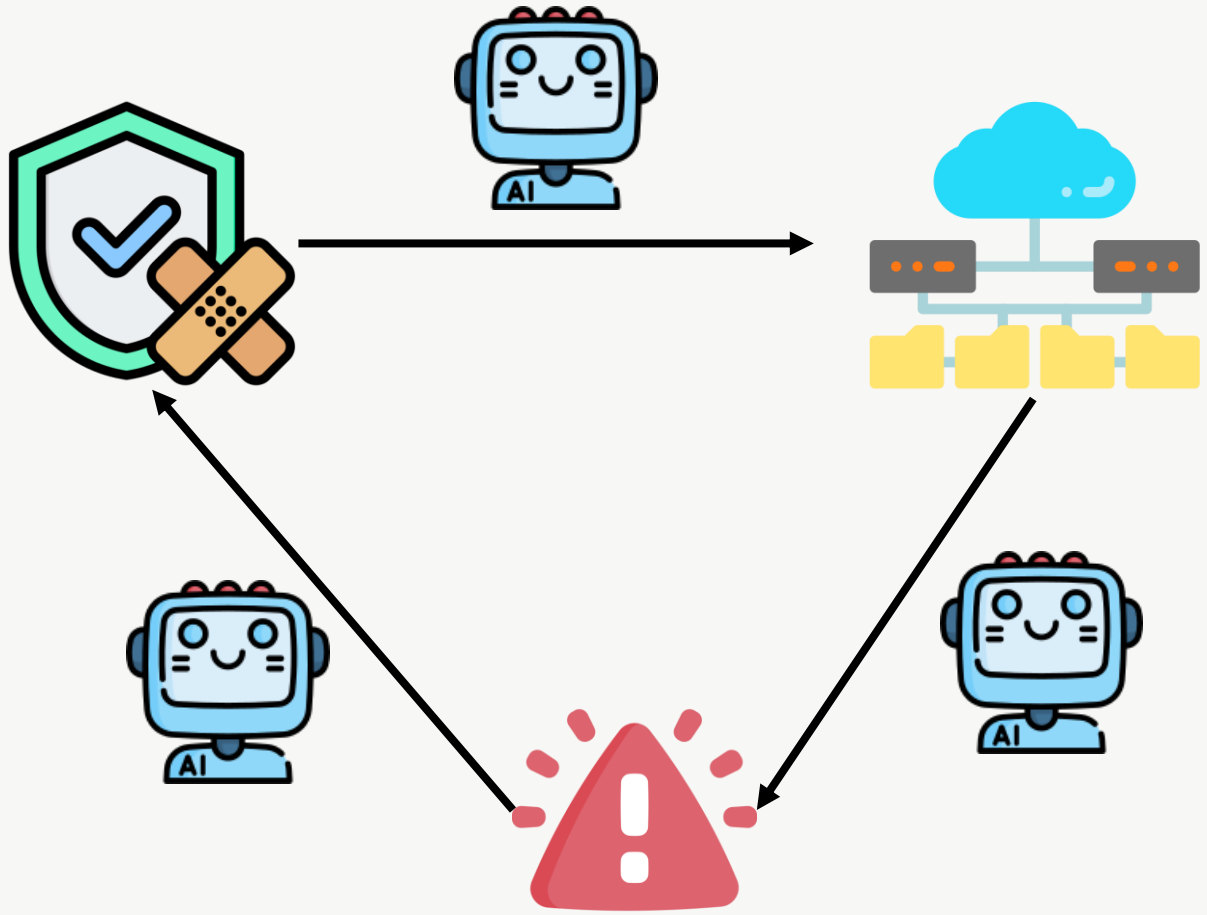
https://arxiv.org/pdf/2507.12472

Use LLMs to Recommend Solutions



https://dl.acm.org/doi/pdf/10.1145/3639476.3639773

A Dream



Discussion

1. Benchmark
Define a problem.

2. Harness
Design a agent harness to make LLMs work good on this problem.

3. Model
Training the LLMs to improve its overall performance.



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Questions

Congratulations! Class is over!